Cymbal Meet — Troubleshooting Guide: Call Quality

Diagnosing and Resolving Audio/Video Quality Issues

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1. Overview

Call quality is the single most important factor in user satisfaction with Cymbal Meet. Poor call quality drives users back to legacy tools, reduces ad-hoc call adoption, and increases support ticket volume. This guide covers call quality issues for all Cymbal Meet clients — desktop, web, mobile, and conference room devices.

Call Quality Score

Every Cymbal Meet call receives a quality score from 1.0 to 5.0, calculated from:

- Audio clarity (30% weight)
- Video resolution stability (25% weight)
- Latency and jitter (20% weight)
- Packet loss impact (15% weight)
- Connection reliability / drop count (10% weight)

Score Range	Rating	[redacted] Experience
4.5-5.0	Excellent	

1. E[illegible]e only O[illegible]dio device is active per participant loca[illegible]
2. U[illegible]eadsets [illegible]dividual participants (eliminates acoust[illegible]ho entirely)
3. C[illegible]rence ro[illegible]evices: verify echo cancellation is enab[illegible]ettings > Audio > Echo C[illegible]llation)
4. I[illegible]o persist[illegible] conference room device, run Audio Ca[illegible]tion: Settings > Audio > Calibrate **[illegible]** Audio
5. C[illegible]

Audio[illegible]

Symp[illegible]

Root C[illegible]

- F[illegible]
- I[illegible]
- C[illegible]
- E[illegible]

Resol[illegible]

1. C[illegible]
2. I[illegible]
3. D[illegible]
4. E[illegible]
5. E[illegible] l[illegible]

2.2 V[illegible]

Low R[illegible]

Symp[illegible]

Root C[illegible]

- E[illegible] c[illegible]
- C[illegible]
- H[illegible]

Bandw[illegible]

R[illegible]
108[illegible]
720[illegible]
480[illegible]
240[illegible]

Resol[illegible]

1. Run ba[illegible]
2. If band[illegible]

3. For desktop: ensure hardware acceleration is ena (Settings > Video > Hardware Acceleratio
4. For web client: use Chrome or Edge (hardware ac ation support). Firefox and Safari may fal back to software encoding.

Video Freezing

Symptoms: Video freezes while audio continues; froze 2-10 seconds then resumes; "slideshow ct.

Root Causes:

- Jitter (variation in packet arrival time) above 30m
- Packet loss on the video stream (UDP)
- Network congestion causing buffer underrun
- Corporate proxy or firewall inspecting/delaying vi ackets

Resolution Steps:

1. Check jitter metrics: Admin Console > Calls > sel ll > Network Metrics > Jitter
2. If jitter >30ms consistently: configure QoS to pric Cymbal Meet video traffic (DSCP AF41)
3. Verify media traffic is NOT routed through a proxy Device Performance guide, Section 5)
4. If the issue is specific to the web client: check br extensions that may interfere with WebRTC
5. Enable Adaptive Jitter Buffer: Settings > Video > ive Buffer (adds 20-50ms latency but smooths playback)

2.3 Call Drops

Symptoms: Participants are disconnected mid-call; "Reconnecting..." banner appears frequently; calls en abruptly.

Root Causes:

- Network connectivity loss (WiFi disconnect, VPN timeo
- Firewall session timeout (TCP idle timeout too short for
- Corporate VPN reconnection cycle (VPN drops and rec
- Client crash due to memory or CPU exhaustion

Understanding Drop Count:

Drops per Call	Assessment	
0	Normal	N/A
1	Occasional issue	Transient network
2-3	Significant issue	Systematic netwo
4+	Critical	Infrastructure fail

Healthy organizations average <0.3 drops per call. An ave requiring immediate investigation.

Resolution Steps:

1. Identify pattern: Admin Console > Calls > filter by drops
    - All users affected → Network infrastructure issue
    - Specific users → Individual client or connectivity i
    - Time-of-day pattern → Congestion or scheduled r

Specific client type → Client bug or platform-specific issue

2. F PN-related drops:

   Configure split tunneling to exclude Cymbal Meet traffic from VPN
   f split tunneling is not possible, increase VPN idle timeout to 300 seconds
   Consider Cymbal Meet's Direct Connect mode (bypasses VPN for media while ping signaling on VPN)

3. F ewall-related drops:

   et TCP idle timeout to at least 300 seconds for Cymbal Meet signaling conne s
   llow UDP 10000-20000 for media (stateful firewall should maintain sessions
   erify NAT session timeout is at least 120 seconds

4. F Fi-related drops:

   nable fast roaming (802.11r) on WiFi access points
   nsure adequate WiFi coverage in all meeting-capable areas
   se 5GHz band exclusively for Cymbal Meet devices (less interference than 2 z)

3. C t-Specific Issues

3.1 D op Client (Windows / macOS)

Comm sues:

- H PU usage during calls with gallery view (10+ participants)
  ix: Limit gallery view to 9 tiles (Settings > Video > Gallery Limit)
- S n sharing causes video quality drop
  ix: Enable hardware-accelerated screen capture (Settings > Screen Share > are apture)
- A device switching mid-call when peripherals connect/disconnect
  ix: Lock audio device (Settings > Audio > Lock Device During Calls)

Minim ystem Requirements:

- C cores, 2.0 GHz
- RAM: 8 GB
- Netwo
- OS: Wi

3.2 Web C

Browser Co

Browse
Chrome 1
Edge 100
Firefox 10
Safari 16+

Web C[redacted] Tro[redacted]ooting:

1. C[redacted] brow[redacted]che and cookies for cymbalmeet.com
2. D[redacted]e br[redacted]extensions (ad blockers and privacy extensions ofte[redacted]erfere with WebRTC)
3. C[redacted] tha[redacted]ra and microphone permissions are granted
4. V[redacted] Web[redacted]s not disabled (chrome://webrtc-internals for diagno[redacted])

3.3 M[redacted]e C[redacted] (

Comm[redacted]ssue[redacted]

- E[redacted]y dr[redacted]ri[redacted]
  [redacted]ix: Reduce [redacted]
- A[redacted]routing co[redacted]
  [redacted]ix: Tap th[redacted]
- F[redacted]uality on [redacted]
  [redacted]ix: Enable [redacted]
  [redacted]andwidth [redacted]

3.4 C[redacted]rence R[redacted]

See th[redacted]icated **Tr[redacted]** device[redacted]bleshootin[redacted]

Quick[redacted]ks:

- [redacted] the devi[redacted]
- C[redacted]
- Run re[redacted]
- Verify [redacted]

4. IT Se[redacted]

4.1 Corpo[redacted]

Problem: M[redacted] Cymbal Mee[redacted] significant la[redacted]

Impact:

- Added [redacted]
- Jitter i[redacted]
- Potent[redacted]
- SSL ins[redacted]

Solution:

1. Config[redacted]
   - `*.cymbalmeet.com`
   - `media.cymbalmeet.com`
   - `signaling.cymbalmeet.com`
2. If full b[redacted]
   (`media.cymbalmeet.com`

[redacted] with the security team to add Cymbal Meet's certificate chain to the SSL inspection allowlist

4.2 [redacted]ed-Down Client Configuration

Pro[redacted] administrators sometimes restrict Cymbal Meet client features for security reasons, ina[redacted]ly disabling functionality that impacts quality and adoption.

Co[redacted] Restricted Features and Their Impact:

[redacted]ted [redacted]re	Security Rationale	Impact on Users	Recommended Alternative
[redacted]l / [redacted]	Prevent unauthorized meetings	Eliminates ad-hoc calls; users can only join scheduled meetings. Dramatically reduces ad-hoc adoption rates (typically to <5%).	Use meeting policies instead — allow quick calls within the organization but require approval for external guests
[redacted]eer [redacted]	Force all media through corporate network	Adds 20-50ms latency; all calls route through media relay	Allow P2P within corporate network; relay only for external participants
[redacted]	Compliance / [redacted]	Users cannot record [redacted]	Enable recording with automatic DLP scanning and [redacted]
Scr[redacted]	[redacted]	[redacted]	[redacted]
Virt[redacted] bac[redacted]	[redacted]	[redacted]	[redacted]

Asses[redacted]

1. [redacted]
2. [redacted]
3. [redacted]
4. [redacted]

4.3 V[redacted]

Probl[redacted]
This a[redacted]

Soluti[redacted]

- [redacted]
- [redacted]

sit.oaembalmeet.com

signaling.cymbalmeet.com [redacted] AD[redacted]

[redacted]4[redacted]

5. Organization-Wide Quality Improvement Plan

When your organization's average call quality score is below 3.5, follow this systematic approach:

Step 1: Diagnose Scope (Day 1-2)

- Pull all quality reports for the past 30 days from Admin Console > Analytics > Call Quality
- Segment by: client type, location/office, time of day, call size
- Identify whether the issue is universal or concentrated

Step 2: Address Infrastructure (Day 3-7)

- Network assessment: Verify QoS, bandwidth, proxy configuration
- Device assessment: Check minimum system requirements, firmware versions
- Refer to the relevant sections of this guide and the Device Performance guide

Step 3

- [redacted]
- [redacted]
- [redacted]

Step 4

- [redacted]
- [redacted]
- [redacted]

Step 5

- [redacted]
- [redacted]
- Include [redacted]

6. Escal[redacted]

Leve[redacted]	
Self-Servi[redacted]	
L1 Suppor[redacted]	support@cymbalmeet.com
L2 Suppo[redacted]	
L3 Engine[redacted]	

Cymbal Mee[redacted]
Customer S[redacted]